

GE 303: Modern Powertrains: Electric, Hybrid, and IC/Aero Engine

Course Code: GE 303

Course Name: Modern Powertrains: Electric, Hybrid, and IC/Aero Engine

L-T-P-C: 2-0-1-3

Prerequisites: None

Intended for: UG 3rd, 4th year, and PG students

Mutual Exclusion: None

Approval: 61st BoA; listed as a discipline elective in the B.Tech General Engineering curriculum approved by the 48th Senate

Preamble

The course provides an overview of the fundamentals of the modern powertrains, which includes, electric vehicles, hybrid electric vehicles, IC Engines and aero engines/turbines. The learning outcomes will be as follows: Fundamentals and architecture of a typical IC Engine; Advancement in IC Engine technology, Turbocharged, GDI, HCCI, and Flex fuel Engines; Operational fundamentals of an Aero Engine; Types of gas turbines and their working principles; Basics for Hybrid electric powertrain design and their transmission system; Conversion techniques from ICE to HEV; Fundamentals of electric powertrain and design principles for electric vehicles.

The course includes a practical part that will include dismantling & assembly of an IC Engine, dismantling & assembly of an electric scooter, performance and emission experiments on a HEV powertrain, design of planetary gear system for HEV transmission, and overview of MIG-29 aero engine.

Course Modules with Quantitative Lecture Hours

- **Unit 1: IC Engine Fundamentals:** Explanation of the engine specification of a typical 2-wheeler and a 4-wheeler engine. Fundamental difference between SI and CI Engines. Correlation of piston movement and air standard cycles for 2 Stroke & and 4 Stroke Engines. Understanding the impact of engine displacement, bore, stroke, and bore-to-stroke ratio. (6 Hours)
- **Unit 2: Advanced IC Engine Concepts:** Combustion chamber design: SI, DISI, IDI, CI, etc. Valve port and manifold design. Understanding swirl, tumble, supercharging, and turbocharging in engines. Engine block design for air cooled and water-cooled engine. Learning various fuel injection systems: PFI, GDI, Carburetor, etc. (6 Hours)
- **Unit 3: Hybrid Electric Powertrains:** Overview of hybrid electric powertrains. Fundamentals of typical HEV powertrains: Parallel hybrid system, series hybrid system, and mixed. Level of hybridization: micro, mild, full, and plug-in HEV. Understanding transmission system of typical HEVs: Honda City HEV, Toyota Hycross HEV, and Yamaha FZ HEV. (6 Hours)
- **Unit 4: Electric Powertrain:** Understanding the basic elements of an electric powertrain. Type of motors and principles to select motors for an electric powertrain. Comparative study of all the batteries used in electric powertrains: their advantage & disadvantage. Various charging principles. Status of Charging infrastructure in India. (6 Hours)
- **Unit 3: Aero Engine Design:** Fundamentals of gas turbines. Type of aero engines: Turbofan, Turboshaft, and Turboprops, their advantages and disadvantages. Fuel injection system for aero engines. Structural elements of aero engines: compressor, combustion chamber, and afterburners, their use and principles. (4 Hours)

Laboratory/Practical/Tutorial Modules: Hands-on Activities (14 Hours)

1. Dismantling and Assembly of a 2 Stroke Engine.
2. Dismantling and Assembly of a 4 Stroke Engine.
3. Dismantling and Assembly of an Electric Scooter.
4. Dismantling and Assembly of an Electric E-Rikshaw.
5. Design & Manufacturing of HEV Planetary Gear System.
6. Performance & Emission testing of a Parallel HEV Powertrain
7. Visual overview of MIG-29 Aero Engine.

Text Books

1. Kevin Hoag & Brian Dondlinger, *Vehicular Engine Design*, Springer.
2. A. Kolchin & V. Demidov, *Design of Automotive Engines*, MIR Publishers.

References

1. Robert Bosch GmbH, *Diesel-Engine Management & Gasoline-Engine Management*.
2. Ehsani Mehrdad, *Modern Electric, Hybrid Electric, and Fuel Cell Vehicles*.
3. Iqbal Husain, *Electric and Hybrid Vehicles: Design Fundamentals*.
4. David T. Pratt, Jack Mattingly, and William H. Heiser, *Aircraft Engine Design*.